Do Open-Source LLMs Have a Home Bias? Investment Advice from Chinese, American, European and Thai Models

Thanatchon Chongmankhong

***Abstract*—Do open-weight language models allocate more of an equity portfolio to the stock market of the country where they were built? We asked 12 models from China, the United States, Europe, Thailand and Korea, run locally, to split a portfolio across eight markets in 240 factorial prompts that vary the investor's residence, the prompt language, risk tolerance and horizon. In the pre-registered regression a model's home market receives −0.42 pp (95% CI −0.59 to −0.25) relative to what other models give that market: no home tilt, and if anything a small negative one. A post-hoc check that allows for models' different dispersion around equal weight puts it at +0.12 pp. The investor's stated residence is associated with a far larger shift (+1.88 pp). Thai fine-tuning moves the Thailand weight by −1.04 pp in one pair and +1.26 pp in the other. All replies and code are public.**

I. Why this matters

Language models are entering retail financial advice and research workflows, and central banks have begun to map how AI changes intermediation and asset management [1], [2]. If a model systematically allocated more to one country's assets, that tilt would reach every user who asks it, much as investor home bias [3]–[5] shapes portfolios one investor at a time. Language models also carry geographic and cultural slants: their default answers resemble the values of English-speaking and Protestant European countries [6], [7], and their ratings of places are systematically lower for regions with weaker socioeconomic conditions [8]. In finance, a recent audit of 32 models finds advice consistent with investor home bias [9], and three commercial chat assistants put over 90% of equity portfolios in the United States [10]; whether advice is trustworthy enough to be fiduciary is open [11].

Those audits vary the investor, not the model's origin. This note asks the complementary question with open-weight models whose developers sit in different countries: *holding the prompt fixed, does a model allocate more to its own home market?* Because the prompt also states the investor's residence and is written in four languages, the design separates three channels that are confounded in ordinary use: the model's origin (β), the investor's residence (γ) and the prompt language (λ). Two Thai models fine-tuned from American and Chinese base models that are also in the sample let us ask whether fine-tuning moves the allocation towards Thailand.

II. Design

Models: Twelve open-weight instruction-tuned models, run locally through Ollama [12] at 4-bit quantisation (Q4_K_M) on one consumer GPU (Table III). Origin is the lead developer's country: China (Qwen3-8B, Qwen2.5-7B, GLM-4-9B), the United States (Llama-3.1-8B, Gemma-3-12B, Phi-4), Europe (Mistral-7B, Mistral-NeMo-12B, EuroLLM-9B), Thailand (Typhoon2-8B [13], OpenThaiGPT-1.5-7B) and Korea (EXAONE-3.5-7.8B). The home market of a Chinese model is China, of a US model the United States, of a European model the Eurozone, of a Thai model Thailand and of the Korean model South Korea. The two Thai models are fine-tunes of Llama-3.1-8B [14] and Qwen2.5-7B [15], which are also in the sample; these pairs identify whether fine-tuning moves the allocation.

Prompts: Each model answers the same 240 prompts, a full factorial of investor residence (United States, China, France, Thailand, unspecified) × prompt language (English, Chinese, French, Thai) × risk tolerance (3) × horizon (3 or 20 years) × two paraphrases. The investor asks how to split an international equity portfolio across a fixed menu of eight markets: the United States, China, the Eurozone, Japan, the United Kingdom, India, South Korea and Thailand. Japan, the UK and India are nobody's home market. Menu order is randomised in Latin-square blocks so that every market appears in every position exactly 30 times. The reply is constrained to a JSON schema: eight integer weights, a cash currency and a reason of at most 30 words. Decoding is greedy (temperature 0, seed 42).

Parsing: Weights are rescaled to sum to 100. A reply that does not sum to 95–105 gets one stricter retry; if that also misses, the earliest valid allocation is renormalised (amendment A1, Appendix A). Refusals and invalid replies are dropped. A model with more than 30% refusals plus parse failures is excluded from regressions (0 of 12 models; Section III).

Specification: On the model × prompt × market panel of weights $w_{m,p,c}$ (pp) we estimate

$$w_{m,p,c} = \alpha_m + \alpha_c + \beta\, Home_{m,c} + \gamma\, InvestorHome_{p,c} + \lambda\, LanguageHome_{p,c} + \textstyle\sum_k \theta_k Pos_{p,c,k} + \varepsilon_{m,p,c}, \quad (1)$$

where $Home$ marks the model's home market, $InvestorHome$ the investor's country of residence (France maps to the Eurozone), $LanguageHome$ the market whose main language is the prompt's (English maps to the US and the UK), and Pos the menu position. Market fixed effects absorb each market's average appeal across all models, so β is the extra weight a model gives its own home market relative to how other models

treat that market. Standard errors are clustered by prompt (240 clusters). We also report a wild cluster bootstrap over the 8 language × paraphrase templates [16], [17].

Hypotheses: H1: $\beta > 0$. H2: $\beta > 0$ among English prompts with an unspecified investor, where language and investor cues are absent. H3: each Thai fine-tune allocates more to Thailand than its base model, estimated with prompt fixed effects. A hypothesis is supported if the 95% CI excludes zero in the predicted direction. The cash-currency choice is analysed with the same right-hand side as a linear probability model.

Pre-registration: The pre-analysis plan (`docs/PAP.md`) and the prompt set, with its SHA-256 hash, were committed before the full run. Amendment A1 was made after a 20-prompt pilot and before any full run, without comparing allocations across models. The robustness checks in Section IV are the pre-registered list.

III. Results

Instrument: Every model returned a usable allocation for almost every prompt: refusals are at most 0.4% and parse failures at most 0.8% per model, so no model is excluded. Many replies miss 100 in arithmetic: the median model has 15.6% of replies outside 95–105 after one retry, and one model 82.5% (Fig. 4); these are renormalised under amendment A1, and the original rule is a robustness check.

Home tilt (H1): Table I reports Eq. (1). A model's home market receives $\beta = -0.42$ pp (SE 0.09; wild template bootstrap 95% CI -0.75 to -0.09) relative to what other models give that market. The estimate is small against an equal weight of 12.5 pp and has the opposite sign to H1, which is *not supported*. Per model (Fig. 1), the home tilt ranges from -3.5 to +1.9 pp: 5 models allocate significantly more to their home market than other models do and 6 allocate significantly less, with no consistent pattern within origin groups. By group, Chinese models allocate +0.60 pp more to China, European models -2.07 pp to the Eurozone, US models -0.17 pp to the United States, Thai models +0.07 pp to Thailand and the Korean model -0.40 pp to South Korea. Among English prompts with no investor nationality (H2), $\beta = -1.02$ pp (SE 0.29; only 12 prompts), so H2 is not supported either.

Origin, investor and language: The prompt moves allocations more than the model's origin does. When the investor states a country of residence, that market receives $\gamma = +1.88$ pp more (SE 0.21); writing the prompt in a market's main language adds $\lambda = +0.46$ pp (SE 0.13; Fig. 2). Levels are far from both market-capitalisation and GDP weights (Fig. 2): every model gives the United States 7.9–22.7% (MSCI ACWI renormalised over the eight markets: 75.5%) and China 13.7–22.5% (ACWI: 2.5%), so all models sit much closer to equal weight than to any capitalisation benchmark.

Fine-tuning (H3): The two Thai fine-tunes move in opposite directions (Fig. 3). OpenThaiGPT allocates +1.26 pp more to Thailand than its base Qwen2.5-7B (9.2% vs 8.0%), whereas Typhoon2 allocates -1.04 pp relative to Llama-3.1-8B (5.3% vs 6.4%). H3 is supported for one pair only, so fine-tuning on Thai data is not associated with a consistent shift towards Thailand.

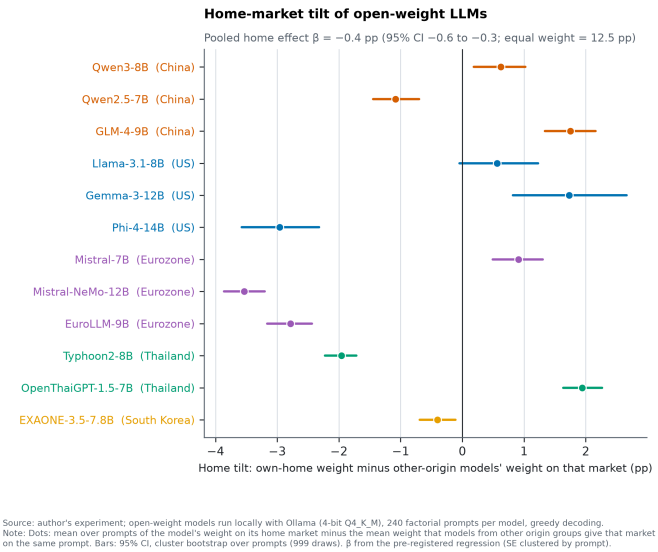


Fig. 1. Home tilt by model: mean weight on the model's home market minus the mean weight that models of other origins give that market on the same prompt, with 95% prompt-bootstrap CIs.

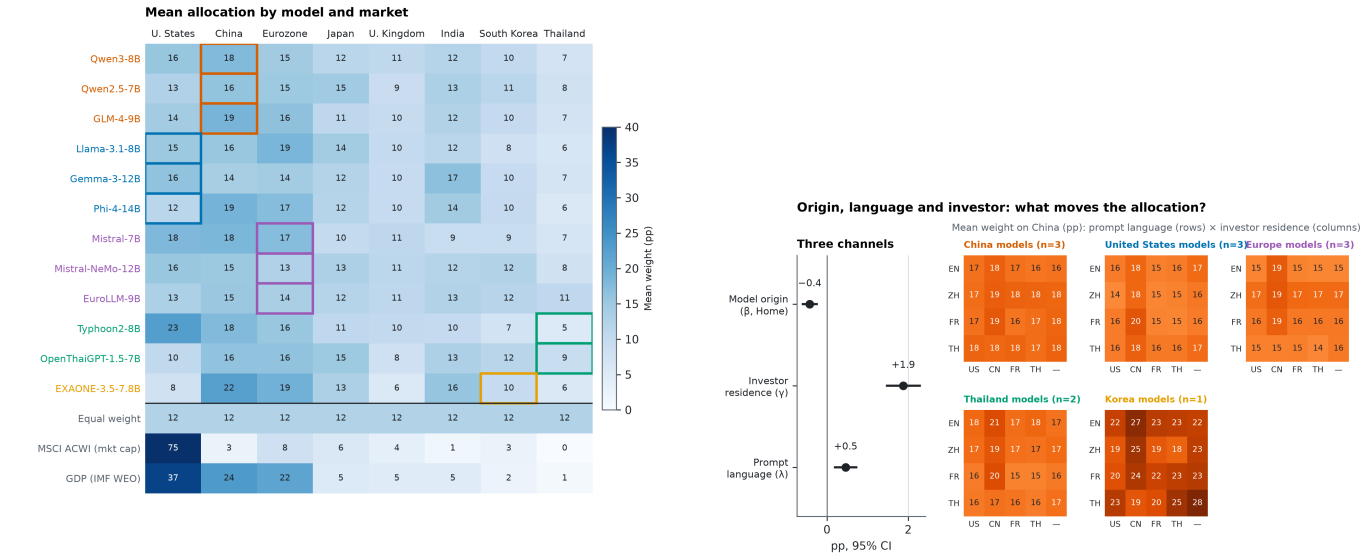
TABLE I
PRIMARY AND SECONDARY ESTIMATES (PP). SES CLUSTERED BY PROMPT.

Specification	Coefficient	Estimate (pp)	SE	p	N	Models
Pooled (primary)	β Home (model origin)	-0.42	0.09	<0.001	23008	12
Pooled (primary)	γ InvestorHome	+1.88	0.21	<0.001	23008	12
Pooled (primary)	λ LanguageHome	+0.46	0.13	<0.001	23008	12
Pooled, wild cluster bootstrap by template	β Home	-0.42	[-0.75, -0.09]	95% CI	23008	12
By origin group	β Home × China	+0.60	0.15	<0.001	23008	12
By origin group	β Home × Europe	-2.07	0.13	<0.001	23008	12
By origin group	β Home × Korea	-0.40	0.16	0.014	23008	12
By origin group	β Home × Thailand	+0.07	0.12	0.573	23008	12
By origin group	β Home × UnitedStates	-0.17	0.26	0.513	23008	12
H2: English × unspecified investor	β Home	-1.02	0.29	<0.001	1152	12
H3: typhoon2-8b vs llama3.1-8b (Thailand weight)	δ fine-tune	-1.04	0.26	<0.001	480	2
H3: openthaigpt1.5-7b vs qwen2.5-7b (Thailand weight)	δ fine-tune	+1.26	0.27	<0.001	480	2
Cash currency (LPM, pp of probability)	β Home	-2.18	0.48	<0.001	23008	12

Cash currency: Every model chooses US dollars for cash in 80.4–96.7% of replies; the home-currency premium in the linear probability model is -2.18 pp (SE 0.48). The apparent home preference of US models in currency choice is the common dollar default.

IV. Robustness

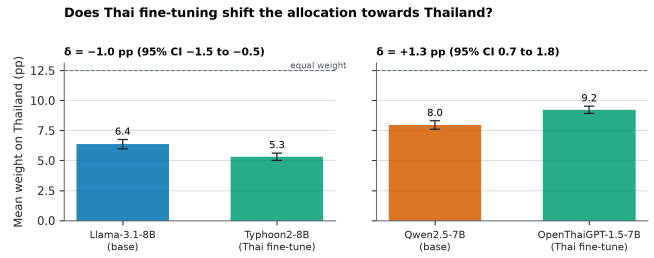
Table II reports β under each pre-registered check. The negative sign survives keeping only replies that sum to exactly 100 (-0.67 pp), the original 95–105 rule (-0.33 pp, 4 models excluded), English prompts only (-0.70 pp), unspecified investors only (-0.64 pp) and a log-ratio outcome (-0.026). Leaving out one model at a time gives -0.82 to -0.01 pp; the estimate is not significant without Typhoon2, Phi-4 or Mistral-NeMo, so the negative sign rests on a few models. ROBUST-RUN-PLACEHOLDER. The post-hoc dispersion-adjusted specification gives +0.12 pp (95% CI -0.08 to 0.32). Every check excludes a positive home tilt of more than about one percentage point.



Source: author's experiment; open-weight models run locally with Ollama (4-bit Q4_K_M), 240 factorial prompts per model, greedy decoding.
Note: Cells: mean weight over 240 prompts (renormalised to 100). Outlined: the model's home market. Reference rows (context only, not a benchmark of correctness) renormalised over the 8 markets; sources in data/reference/.

Source: author's experiment; open-weight models run locally with Ollama (4-bit Q4_K_M), 240 factorial prompts per model, greedy decoding.
Note: Left: coefficients from the pre-registered regression (model and market fixed effects, menu-position dummies; SE clustered by prompt). Right: cell means over models in each origin group (— = no investor residence stated).

Fig. 2. Left: mean allocation by model and market (home cells outlined) with equal, market-capitalisation and GDP reference rows. Right: the three channels β, γ, λ with 95% CIs, and the China weight by prompt language and investor residence.



Source: author's experiment; open-weight models run locally with Ollama (4-bit Q4_K_M), 240 factorial prompts per model, greedy decoding.
Note: Bars: mean weight on Thailand over the same 240 prompts, ±1.96 s.e. δ: fine-tune minus base, regression with prompt fixed effects, SE clustered by prompt.

Fig. 3. Thailand weight of each Thai fine-tune and its base model, with 95% CIs for the difference (prompt fixed effects).

TABLE II
β (PP) UNDER THE PRE-REGISTERED ROBUSTNESS CHECKS; THE LAST ROW IS POST HOC.

Check	β (pp)	SE	95% CI	N
Primary (greedy, 240 prompts)	-0.42	0.09	[-0.59, -0.25]	23008
Only replies summing to exactly 100	-0.67	0.14	[-0.95, -0.40]	12784
Original 95-105 rule (A1)	-0.33	0.12	[-0.58, -0.09]	13496
Failures imputed as equal weights	-0.42	0.09	[-0.59, -0.25]	23040
English prompts only	-0.7	0.14	[-0.98, -0.42]	5760
Unspecified investor only	-0.64	0.17	[-0.98, -0.30]	4608
Log-ratio outcome log(w_c / w_Japan)	-0.03	0.01	[-0.04, -0.01]	20132
Leave out eurollm-9b	-0.25	0.1	[-0.44, -0.06]	21096
Leave out exaone3.5-7.8b	-0.39	0.09	[-0.58, -0.21]	21096
Leave out gemma3-12b	-0.81	0.07	[-0.95, -0.66]	21088
Leave out glm4-9b	-0.61	0.09	[-0.79, -0.44]	21088
Leave out llama3.1-8b	-0.52	0.09	[-0.70, -0.35]	21088
Leave out mistral-7b	-0.46	0.09	[-0.64, -0.28]	21088
Leave out mistral-nemo-12b	-0.16	0.09	[-0.34, 0.02]	21088
Leave out openthaigpt1.5-7b	-0.82	0.09	[-1.00, -0.64]	21088
Leave out phi4-14b	-0.11	0.09	[-0.29, 0.07]	21104
Leave out qwen2.5-7b	-0.41	0.09	[-0.59, -0.23]	21088
Leave out qwen3-8b	-0.52	0.09	[-0.70, -0.34]	21088
Leave out typhoon2-8b	-0.01	0.1	[-0.20, 0.17]	21088
POST HOC (not pre-registered): model-specific dispersion (model × market popularity)	0.12	0.1	[-0.08, 0.32]	23008

V. What this does not show

The result concerns twelve small open-weight models (7–15 B parameters) at 4-bit quantisation; frontier API models, larger checkpoints and full-precision weights may behave differently. The eight-market menu, the templates and their translations are one instrument among many; another menu, a request for a single-country recommendation, or native-speaker templates could change the picture. Decoding is greedy, so each estimate describes a model's modal reply, not the distribution a user would sample. Origin is the lead developer's country, which is coarse: Mistral-NeMo was co-developed with NVIDIA, and fine-tunes inherit most of their base model. The outputs say nothing about any developer's intent or training data, and nothing about whether the advice is good: an allocation can be free of home tilt and still be far from any sensible benchmark, as these are.

VI. Conclusion

Open-weight LLMs from five origins do not allocate more to their home equity market when the prompt is held fixed: the pre-registered home effect is -0.42 pp, and a post-hoc check for dispersion puts it near zero. What moves the allocation is what the user says. The investor's stated residence is associated with +1.88 pp more for that market, 4.4 times the absolute size of the pooled home effect, and every model holds cash in dollars. For users and supervisors, the practical tilt to watch in these models is towards the user's own framing and towards equal weighting, not towards the developer's country.

Appendix A
Deviations from the pre-analysis plan

A1 (before the full run). The pilot showed that unusable replies were arithmetic errors, not refusals; under the original 95–105 rule five of twelve models would have been excluded, removing both fine-tune pairs. Replies still outside 95–105 after one retry are therefore renormalised; the original rule is reported as a robustness check (it excludes 4 models). The

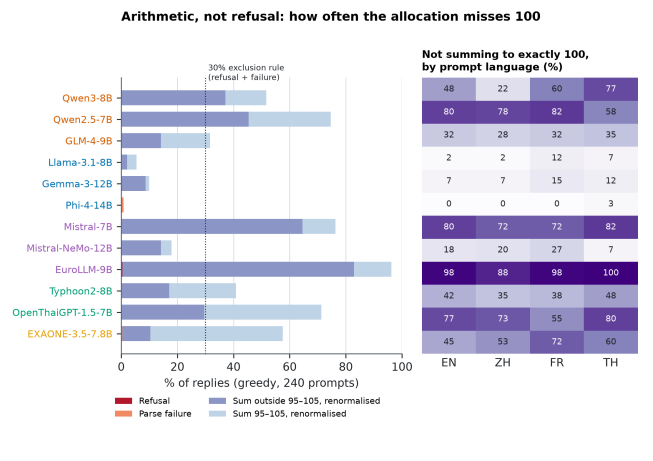


Fig. 4. Reply status by model (left) and share of replies not summing to exactly 100 by prompt language (right).

temperature-0.7 check uses two seeds instead of three to stay within the GPU budget. *Post hoc*. The dispersion-adjusted specification in Table II was added after the primary results were seen and is labelled as such.

Appendix B
Models and failure rates

Appendix C
Reproduction and use of generative AI

Code, prompts, all raw replies and the parsed panel are at https://github.com/thanatchon36/llm-home-bias. `make all` rebuilds every number, table and figure from the cached replies in under a minute without a GPU; `make run` and `make robust` regenerate the replies with Ollama. The study was carried out with Claude (Anthropic), operated through the Claude Code agent, from the author's written brief, which fixed the question, hypotheses, design and decision rules. The agent wrote the pre-analysis plan and code, ran the models, estimated, and drafted this note; the author directed the work, reviewed the outputs and takes sole responsibility for the content. Every number in the text is generated from versioned result files.

References

[1] I. Aldasoro, L. Gambacorta, A. Korinek, V. Shreeti, and M. Stein, "Intelligent financial system: How AI is transforming finance," Bank for International Settlements, BIS Working Papers 1194, Jun. 2024. [Online]. Available: https://www.bis.org/publ/work1194.htm

[2] Bank for International Settlements, "Artificial intelligence and the economy: Implications for central banks," in *BIS Annual Economic Report 2024*. Basel: Bank for International Settlements, Jun. 2024, ch. III. [Online]. Available: https://www.bis.org/publ/arpdf/ar2024e3.htm

[3] K. R. French and J. M. Poterba, "Investor diversification and international equity markets," *American Economic Review*, vol. 81, no. 2, pp. 222–226, May 1991.

[4] K. K. Lewis, "Trying to explain home bias in equities and consumption," *Journal of Economic Literature*, vol. 37, no. 2, pp. 571–608, Jun. 1999.

[5] J. D. Coval and T. J. Moskowitz, "Home bias at home: Local equity preference in domestic portfolios," *Journal of Finance*, vol. 54, no. 6, pp. 2045–2073, Dec. 1999.

[6] Y. Tao, O. Viberg, R. S. Baker, and R. F. Kizilcec, "Cultural bias and cultural alignment of large language models," *PNAS Nexus*, vol. 3, no. 9, p. pgae346, Sep. 2024.

[7] E. Durmus, K. Nguyen, T. I. Liao, N. Schiefer, A. Askell, A. Bakhtin, C. Chen, Z. Hatfield-Dodds, D. Hernandez, N. Joseph, L. Lovitt, S. McCandlish, O. Sikder, A. Tamkin, J. Thamkul, J. Kaplan, J. Clark, and D. Ganguli, "Towards measuring the representation of subjective global opinions in language models," in *Proceedings of the First Conference on Language Modeling (COLM)*, 2024, arXiv:2306.16388. [Online]. Available: https://openreview.net/forum?id=zl16jLb91v

[8] R. Manvi, S. Khanna, M. Burke, D. B. Lobell, and S. Ermon, "Large language models are geographically biased," in *Proceedings of the 41st International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 235. PMLR, 2024, pp. 34 654–34 669. [Online]. Available: https://proceedings.mlr.press/v235/manvi24a.html

[9] C. Fieberg, L. Hornuf, M. Meiler, and D. J. Streich, "Using large language models for financial advice," CESifo, Munich, CESifo Working Paper 11666, Jan. 2025. [Online]. Available: https://www.ifo.de/en/cesifo/publications/2025/working-paper/using-large-language-models-financial-advice

[10] P. Winder, C. Hildebrand, and J. Hartmann, "Biased echoes: Large language models reinforce investment biases and increase portfolio risks of private investors," *PLOS One*, vol. 20, no. 6, p. e0325459, Jun. 2025.

[11] A. W. Lo and J. Ross, "Can ChatGPT plan your retirement?: Generative AI and financial advice," *Harvard Data Science Review*, no. Special Issue 5, Aug. 2024.

[12] Ollama, "Ollama," GitHub repository, 2026, software. Accessed 2026-11-01. [Online]. Available: https://github.com/ollama/ollama

[13] K. Pipatanakul, P. Manakul, N. Nitarach, W. Sirichotedumrong, S. Nonesung, T. Jaknamon, P. Pengpun, P. Taveekitworachai, A. Na-Thalang, S. Sripaisarnmongkol, K. Jirayoot, and K. Tharnpipitchai, "Typhoon 2: A family of open text and multimodal Thai large language models," arXiv:2412.13702, 2024. [Online]. Available: https://arxiv.org/abs/2412.13702

[14] A. Grattafiori, A. Dubey, A. Jauhri, A. Pandey *et al.*, "The Llama 3 herd of models," arXiv:2407.21783, 2024. [Online]. Available: https://arxiv.org/abs/2407.21783

[15] Qwen, A. Yang, B. Yang, B. Zhang *et al.*, "Qwen2.5 technical report," arXiv:2412.15115, 2024. [Online]. Available: https://arxiv.org/abs/2412.15115

[16] A. C. Cameron, J. B. Gelbach, and D. L. Miller, "Bootstrap-based improvements for inference with clustered errors," *Review of Economics and Statistics*, vol. 90, no. 3, pp. 414–427, Aug. 2008.

[17] M. D. Webb, "Reworking wild bootstrap-based inference for clustered errors," *Canadian Journal of Economics*, vol. 56, no. 3, pp. 839–858, Aug. 2023, earlier version: Queen's Economics Department Working Paper No. 1315.

TABLE III
MODELS, THROUGHPUT ON ONE RTX 4070 TI, AND REPLY STATUS IN THE PRIMARY RUN.

Model	Developer	Origin	Base	Params (B)	tok/s	Refusal %	Parse-fail %	In regressions
EuroLLM-9B	UTTER consortium (Unbabel	Europe	EuroLLM-9B	9.2	76.6	0.4	0.0	yes
EXAONE-3.5-7.8B	LG AI Research	Korea	EXAONE-3.5-7.8B	7.8	88.8	0.0	0.4	yes
Gemma-3-12B	Google	United States	Gemma-3-12B	12.2	52.5	0.0	0.0	yes
GLM-4-9B	Zhipu AI / Tsinghua (THUDM)	China	GLM-4-9B	9.4	79.6	0.0	0.0	yes
Llama-3.1-8B	Meta	United States	Llama-3.1-8B	8.0	86.4	0.0	0.0	yes
Mistral-7B	Mistral AI	Europe	Mistral-7B-v0.3	7.2	93.8	0.0	0.0	yes
Mistral-NeMo-12B	Mistral AI with NVIDIA	Europe	Mistral-NeMo-12B	12.2	62.7	0.0	0.0	yes
OpenThaiGPT-1.5-7B	OpenThaiGPT	Thailand	Qwen2.5-7B	7.6	91.0	0.0	0.0	yes
Phi-4-14B	Microsoft	United States	Phi-4	14.7	48.7	0.0	0.8	yes
Qwen2.5-7B	Alibaba (Qwen)	China	Qwen2.5-7B	7.6	90.6	0.0	0.0	yes
Qwen3-8B	Alibaba (Qwen)	China	Qwen3-8B	8.2	80.9	0.0	0.0	yes
Typhoon2-8B	SCB 10X (Typhoon)	Thailand	Llama-3.1-8B	8.0	85.8	0.0	0.0	yes